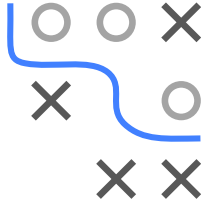
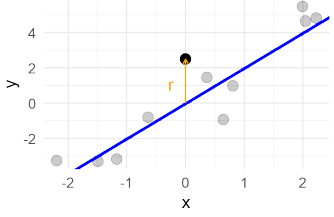
Supervised Learning



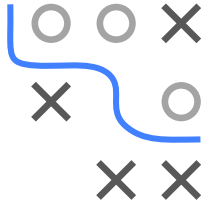
Advanced Risk Minimization
Pseudo-Residuals



Learning goals

- Concept of pseudo-residuals
- PRs for common losses

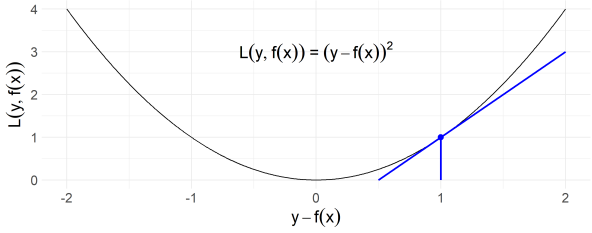
PSEUDO-RESIDUALS



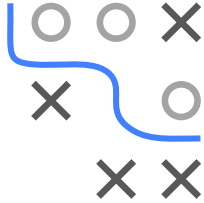
- In regression, residuals are defined as $r(\boldsymbol{x}) := y - f(\boldsymbol{x})$
- Generalize concept to **pseudo-residuals**:

$$\tilde{r}(\boldsymbol{x}) := -\frac{dL\left(y, f(\boldsymbol{x})\right)}{df(\boldsymbol{x})}$$

- If we wiggle $f(\boldsymbol{x})$, how much does L change?
- Can be used for score-based classifiers and other models
- Note that $\tilde{r}(\boldsymbol{x})$ depends on y, $f(\boldsymbol{x})$ and L

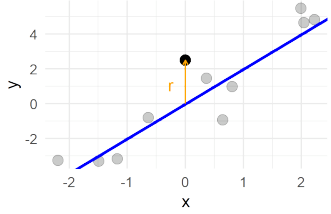


BEST POINT-WISE UPDATE

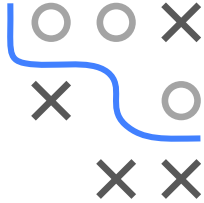


- Assume we have (partially) fitted a model $f(\boldsymbol{x})$ to data $\mathcal{D}$
- Assume we could update $f(\boldsymbol{x})$ point-wise as we like
- Under squared loss, for a fixed $\boldsymbol{x} \in \mathcal{X}$, the best point-wise update is the direction of the residual $r(\boldsymbol{x}) = y - f(\boldsymbol{x})$

$$f(\boldsymbol{x}) \leftarrow f(\boldsymbol{x}) + r(\boldsymbol{x})$$



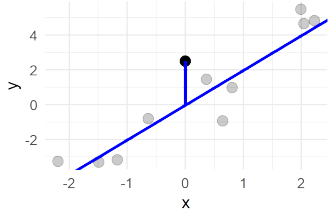
BEST POINT-WISE UPDATE



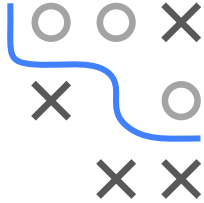
- Assume we have (partially) fitted a model $f(\boldsymbol{x})$ to data $\mathcal{D}$
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- Under squared loss, for a fixed $\boldsymbol{x} \in \mathcal{X}$, the best point-wise update is the direction of the residual $r(\boldsymbol{x}) = y - f(\boldsymbol{x})$

$$f(\boldsymbol{x}) \leftarrow f(\boldsymbol{x}) + r(\boldsymbol{x})$$

- Point-wise error at this specific $\boldsymbol{x}$ becomes 0



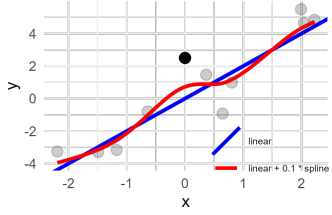
BEST POINT-WISE UPDATE



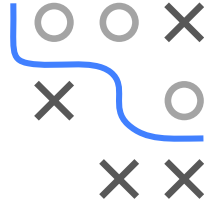
- Assume we have (partially) fitted a model $f(\boldsymbol{x})$ to data $\mathcal{D}$
- Assume we could update $f(\boldsymbol{x})$ point-wise as we like
- Under squared loss, for a fixed $\boldsymbol{x} \in \mathcal{X}$, the best point-wise update is the direction of the residual $r(\boldsymbol{x}) = y - f(\boldsymbol{x})$

$$f(\boldsymbol{x}) \leftarrow f(\boldsymbol{x}) + r(\boldsymbol{x})$$

- (In gradient boosting, which we cover later, we don't do point-wise updates but "smoothly distort" f so we generalize)



APPROXIMATE BEST POINT-WISE UPDATE



- Best local change of f at $\boldsymbol{x}$ to reduce loss most:

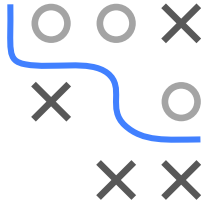
$$f(\boldsymbol{x}) \leftarrow f(\boldsymbol{x}) - \frac{dL\left(y, f(\boldsymbol{x})\right)}{df(\boldsymbol{x})}$$

- This is effectively the PR

$$f(\boldsymbol{x}) \leftarrow f(\boldsymbol{x}) + \tilde{r}(\boldsymbol{x})$$

- (Such iterative updates of f like a loss-reducing GD in function space is the major underlying idea of GB)

GD IN ML AND PSEUDO-RESIDUALS



- In GD, we move in the direction of the negative gradient by updating the parameters:

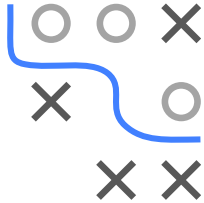
$$\boldsymbol{\theta}^{[t+1]} = \boldsymbol{\theta}^{[t]} - \alpha^{[t]} \cdot \nabla_{\boldsymbol{\theta}} \, \mathcal{R}_{\text{emp}}(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\boldsymbol{\theta}^{[t]}}$$

- Using the chain rule:

$$\begin{aligned}
\nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) &= \sum_{i=1}^{n} \left. \frac{dL(y^{(i)}, f(\boldsymbol{x}))}{df(\boldsymbol{x})} \right|_{f=f\left(\boldsymbol{x}^{(i)} \mid \boldsymbol{\theta}\right)} \cdot \nabla_{\boldsymbol{\theta}} f\left(\boldsymbol{x}^{(i)} \mid \boldsymbol{\theta}\right) \\
&= -\sum_{i=1}^{n} \tilde{r}^{(i)} \cdot \nabla_{\boldsymbol{\theta}} f\left(\boldsymbol{x}^{(i)} \mid \boldsymbol{\theta}\right).
\end{aligned}$$

- Update is loss-optimal directional change of model output and a loss-independent derivative of $f(\boldsymbol{x})$
- This is a flexible, nearly loss-independent variant of GD

PSEUDO-RESIDUALS FOR COMMON LOSSES



Loss	**Domain of y**	**Pseudo residual $\tilde{r}$**
Squared loss	$y \in \mathbb{R}$	$y - f(\boldsymbol{x})$
Bernoulli loss	$y \in \{0, 1\}$	$y - s(f(\boldsymbol{x})) = y - \pi(\boldsymbol{x})$
Multinomial loss	$y \in \{1, \ldots, g\}$	$\mathbb{1}_{\{y=k\}} - \pi_k(\boldsymbol{x})$
Exponential loss	$y \in \{-1, 1\}$	$y \exp(-yf(\boldsymbol{x}))$

NB: $\pi(\boldsymbol{x}) = s(f(\boldsymbol{x})) = \frac{\exp(f(\boldsymbol{x}))}{1+\exp(f(\boldsymbol{x}))}$ is the (sigmoidal) logistic function, and $\pi_k(\boldsymbol{x})$ its multi-class generalization, the softmax